

The Verification Gap in Networked Physical AI: A Post-Semantic Communication Framework

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Abstract—A task-effective proposal is not yet a justified physical action. In networked Physical AI, a proposal may be understood while valid, timely, proposal-bound evidence or the authority required to finalize an action remains unavailable. We call this mismatch the *verification gap* and introduce a *Post-Semantic Communication Framework* for the systems interface between proposal formation and physical execution. The framework begins with application-declared evidence requirements, represents qualifying observations as evidence records, validates supporting and conflicting records through one path, and separates evidence sufficiency from authorized finalization and a downstream runtime gate. It further distinguishes *evidence transfer*, which can enlarge the record set reachable by a finalizer, from *evidence coordination*, which can suppress transmission around records already held at the finalization endpoint. Finite-state framework checks verify that the evaluator implements the declared distinctions consistently. A controlled communication study then compares One-way and Feedback while holding the evidence requirements, finalizer, and evidence-quality assumptions fixed. The selected interaction changes across the declared timing regimes as feedback benefit is weighed against loss, latency, freshness, and deadline cost. Finally, an episode-level reporting schema exposes the common denominators needed to compare such designs and prepares the framework for future measured Physical-AI studies.

Index Terms—Post-semantic communication, verification gap, Networked Physical AI, evidence contracts, evidence records, receiver feedback, authorized finalization.

I. INTRODUCTION

“The aisle appears clear. Proposal: Proceed.” A camera-equipped warehouse robot sends this proposal to another robot approaching an occluded aisle. The bits arrive, the meaning is decoded, and the proposal may be useful. The receiving system

nevertheless cannot justify motion if the application requires a fresh Visual-clearance record and a Radio-no-motion record bound to this proposal, but only one is available. A nearby agent may hold the missing record; it may instead hold an admissible conflict that should stop the action. A task-effective proposal is not yet a justified physical action.

The same distinction arises when a cooperative vehicle receives a trajectory, a manipulator receives a grasp proposal, or an inspection drone receives a route. Networked Physical AI inherits long-standing cyber-physical and cloud-robotics concerns: sensing, computation, communication, and actuation are distributed across components and time [1], [2]. This distribution can also separate evidence location from finalization authority. We define the *verification gap* as the mismatch between a task-effective proposal and the valid, timely, proposal-bound evidence and authorization required at the permitted finalizer before action. Figure 1 places this gap inside a Networked Physical-AI loop [3], [4].

Post-semantic communication names the systems interface after semantic or task-level proposal formation and before physical execution. At this interface, agents expose evidence holdings, transfer validated evidence records, re-evaluate application-declared requirements, and make a sufficient record set available to the authorized finalizer. The interface does not prescribe a transport, topology, learned representation, or number of agents. It identifies a responsibility boundary that is easily hidden inside a generic “receiver decision.” The downstream runtime gate remains independent: it may stop an action even after evidence verification and authorized finalization.

This paper makes four contributions. First, it defines the verification gap as a communication-relevant state between

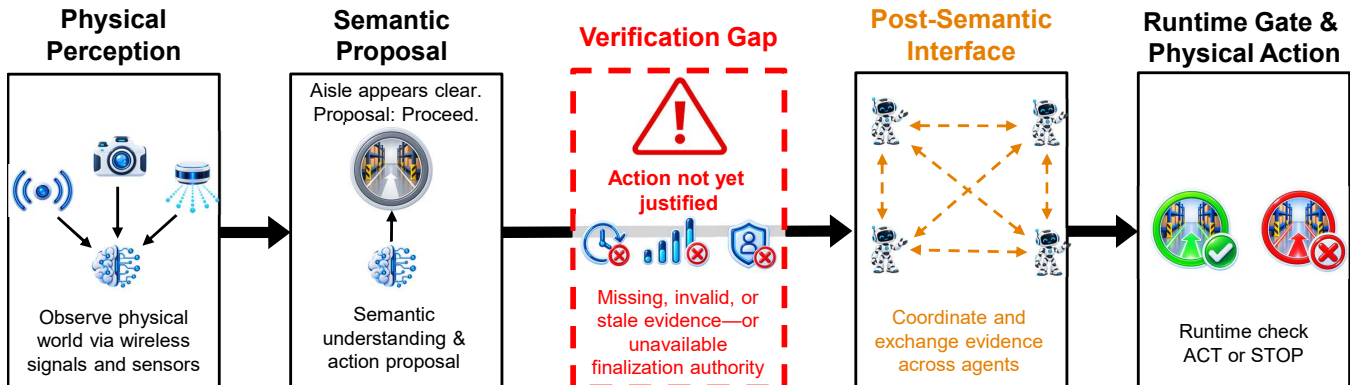


Fig. 1. The verification gap and post-semantic interface within a networked Physical-AI loop. Figure 1 locates the interface between semantic proposal formation and runtime action; Figure 2 defines its internal framework.

proposal understanding and physical action. Second, it proposes the Post-Semantic Communication Framework, which connects evidence requirements, evidence records, verification, authorized finalization, and a downstream runtime boundary. Third, it implements the framework in a finite-state evaluator and checks consistency with the declared structural distinctions, including conflict-first verification and the separation between evidence transfer and evidence coordination. Fourth, a controlled communication study demonstrates a requirement-preserving comparison and supplies reporting guidelines and an episode-level reporting schema for reuse. The checks stop at authorized finalization; measured sensing, credential operation, runtime control, and physical safety remain future work.

II. RELATED WORK AND MOTIVATION

Semantic and task-oriented communication move the objective beyond symbol fidelity toward meaning and task effectiveness [5], [6]. Our question begins after a meaningful proposal already exists. In Weaver’s broad effectiveness sense, the framework isolates the post-proposal transition from task effectiveness to evidence-backed, authorized action toward Level C [7]; it does not redefine communication theory. Receiver side information and interactive coding explain why a sender need not retransmit what a receiver already has [8], [9], while learned multi-agent communication asks when agents should exchange information [10]. These lines motivate efficient delivery, but do not by themselves declare which evidence must be verified before a particular physical action is finalized.

Active perception and information-directed sensing ask what observation to acquire next [11], [12]; distributed sensor fusion asks how heterogeneous observations should be combined [13]. Distributed detection, distributed hypothesis testing, and team decision theory show how observations and decisions can be distributed [14], [15]. Work on information structures in distributed control similarly shows that who knows what, and when, constrains achievable decisions. The present framework fixes the required evidence and authorized finalizer before comparing how already available records reach that endpoint. Physical evidence location is therefore not silently credited as authority over an action.

Embodied multimodal models ground sensors in language, while closed-loop LLM planners use scene descriptions, success signals, and human feedback to replan [16], [17]. Uncertainty-aware planners can ask supervisors for help under ambiguity, while interactive multimodal systems can choose epistemic actions to gather visual, acoustic, haptic, or proprioceptive observations [18], [19]. They motivate AI-assisted diagnosis and information seeking, but do not themselves define which observation qualifies as action-specific evidence, how it is bound to a proposal, or who may authorize finalization. The proposed framework supplies that verification and authority boundary.

Selective prediction supplies accept, reject, and abstain outcomes [20]. Provenance and verifiable-sensing work motivate checkable records [21]–[23]; safety and assurance cases motivate explicit claim–evidence structure [24]; runtime assurance protects execution after a decision [25]; and zero-trust

and capability-based mechanisms inform delegated command authorization [26], [27]. These mechanisms address different layers. A valid record does not grant action authority, and authorized finalization does not replace current-state runtime enforcement.

Contract-based design and assume–guarantee reasoning associate components with environmental assumptions and behavioral guarantees, supporting compositional design, refinement, and verification [28]. Such contracts support compositional verification of modular robots and ROS nodes [29]. Our evidence contract instead declares action-specific claims and validity conditions for proposal-bound records at the permitted finalizer after proposal formation; it does not specify component behavior under environmental assumptions. They are complementary: component contracts constrain behavior; an evidence contract governs what information may justify an action.

Our contribution is to make the post-proposal state an explicit communication framework. It binds application-declared evidence requirements to proposal-specific records, separates verification from authorized finalization and runtime enforcement, and exposes evidence transfer and evidence coordination as distinct communication functions. Related fields contain individual elements of this design; the framework integrates them around the verification gap and a common episode boundary.

Table I is a representational counterexample, not an impossibility theorem: a richer proposal could carry these fields. The point is to preserve them as explicit, independently reportable framework objects.

III. POST-SEMANTIC COMMUNICATION FRAMEWORK

Section II identified episode distinctions that a proposal-only representation can hide. We now define the proposed framework that makes those distinctions explicit and communicable; this is the paper’s main contribution. The framework answers four questions: what evidence is required; how records are validated and combined; who may finalize and what remains for runtime control; and how communication can transfer evidence or coordinate delivery. Figure 2 summarizes the resulting architecture.

A. Evidence Requirements and Records

An *evidence contract* is the application-declared set of evidence requirements and validity rules that must be satisfied before an action can be finalized. It specifies the required claims, acceptable record modalities and sources, support and conflict semantics, freshness and uncertainty limits, provenance checks, proposal binding, and evidence deadline. We use *evidence requirements* for the individual conditions and *evidence contract* for the formal collection of those conditions. A communication policy may change delivery but must not silently relax this contract.

An *evidence record* identifies its claim, relation to that claim, modality, source, measurement time, freshness limit, provenance, uncertainty, proposal binding, and payload or content hash. These fields state which contract item the

TABLE I
SAME PROPOSAL, DIFFERENT ACTION OUTCOMES.

Changed condition	Proposal-only view	Evidence-aware view	Required response
Proposal binding	“Proceed” is unchanged.	The record hash refers to an earlier proposal.	Abstain: the record cannot satisfy the current requirements.
Freshness / expiry	“Proceed” is unchanged.	A required record expires before finalization.	Abstain: evidence is no longer timely.
Admissible conflict	“Proceed” is unchanged.	A validated record conflicts with a required claim.	Reject before testing support completeness.
Finalization authority	“Proceed” is unchanged.	The endpoint holding sufficient records is not authorized to finalize.	Make the required evidence available to the authorized finalizer, or abstain.

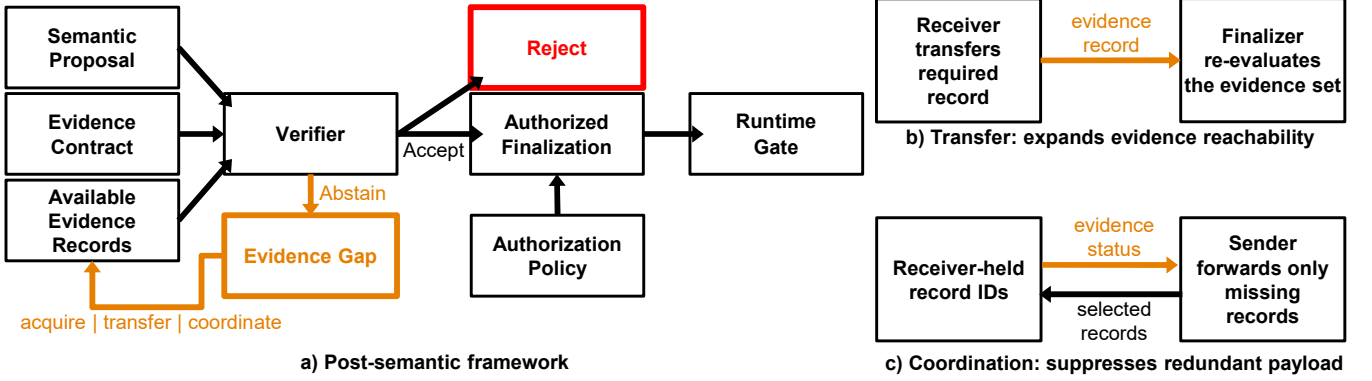


Fig. 2. The Post-Semantic Communication Framework. (a) A semantic proposal, an application-declared evidence contract, and available evidence records enter a common verifier. An admissible conflict yields *Reject*; complete support with no admissible conflict yields *Accept*, which can be finalized only under the separate authorization policy; missing, invalid, stale, or incomplete evidence yields *Abstain* and an evidence gap. The gap may trigger acquisition, transfer, or coordination followed by re-verification, while the runtime gate remains downstream of authorized finalization and outside the present evaluation. (b) Evidence transfer moves a required record to the finalizer, expanding evidence reachability. (c) Evidence coordination exchanges record-status information so that the sender forwards only missing records, suppressing redundant payload; status is not evidence. Acquisition is a future extension, whereas Section IV evaluates transfer and coordination.

record can satisfy and allow a verifier to check the record mechanically. They do not define a new data type: an image, radio observation, signed state estimate, or reference can be an evidence record when it carries the required attributes. In the warehouse example, one record addresses corridor clearance and another addresses the absence of a moving obstacle.

Proposal binding prevents a record for an earlier route or control cycle from satisfying the current requirements. Decision-time freshness prevents a record that was valid when transmission began from being accepted after expiry. Provenance, source, calibration, and uncertainty rules determine whether the application admits the record. A delivery policy may change packetization, routing, references, or interaction without changing any of these tests.

B. Evidence Verification

Supporting and conflicting records traverse the same validation pipeline. Expired, untrusted, calibration-revoked, excessively uncertain, or binding-mismatched records are inadmissible: they can neither satisfy an evidence requirement nor veto a proposal. This common path prevents an invalid negative observation from bypassing the checks applied to support.

The verifier follows the principle *Conflict First, Then Sufficiency*. Among admissible records, any conflict produces *Reject*; a complete supporting set with no conflict produces *Accept*; and every remaining case produces *Abstain*. *Reject* reports a contradiction. *Abstain* reports that the declared evidence requirements were not completed. A conditional record error can still create incorrect support, so verification establishes compliance with the declared contract rather than physical truth.

An unresolved verification gap takes one of two forms: an *evidence gap*, when required admissible records are missing, invalid, stale, incomplete, or unavailable at the permitted finalizer; or an *authority gap*, when evidence is sufficient but authorized finalization is unavailable. An admissible conflict is not a gap: it is a resolved negative outcome that produces *Reject*. An inadmissible conflict does not enter the verifier’s admissible set and cannot veto the proposal.

C. Finalization and Runtime Safety

Evidence sufficiency, authorized finalization, and runtime safety are three separate responsibilities. Evidence sufficiency is the verifier’s result over admissible records. An *authorization policy* is a separate application declaration that identifies

the endpoint or endpoint set permitted to turn an Accept result into an accountable action decision. Authorized finalization requires both verifier Accept and membership in that declared set; authority is not part of the evidence contract. Runtime safety is the later decision of an independent gate using current physical conditions.

An agent can contribute a record without acquiring permission to act. An authorized finalizer can use remote records without claiming to have measured them locally. If sufficient records are assembled at an unauthorized endpoint, the records must become available to the declared finalizer or the system must abstain. A content-addressed manifest may preserve record-set identity across that handoff, but it creates neither authority nor freshness.

The runtime gate receives a finalized action and may return ACT or STOP. It can account for conditions not represented in the evidence contract, including a new obstacle or degraded actuator. Section IV evaluates only through authorized finalization. It does not evaluate runtime-gate performance, sensor-statement correctness, operational credential enforcement, or physical safety.

D. Evidence Transfer and Coordination

Post-semantic communication can serve two functions. *Evidence transfer* carries an admissible record that the authorized finalizer does not yet hold. For sender finalization, successful feedback may return a receiver-held record and thereby enlarge the admissible set reachable before finalization. *Evidence coordination* reports which records are already held at the receiver so that the sender can avoid transmitting redundant payload when the receiver finalizes. The status message guides delivery but does not satisfy an evidence requirement.

The controlled evaluator compares the formal policies *Sender/One-way*, *Sender/Feedback*, *Receiver/One-way*, and *Receiver/Feedback*. The first term fixes the authorized finalizer; the second changes interaction. A One-way policy sends all validated sender-held records. Feedback adds a reverse logical message. Under receiver finalization, loss of status triggers the declared One-way fallback, so coordination does not acquire an artificial reachability advantage.

Both functions have costs. A reverse message can be lost, add fixed latency, age a record past its lifetime, or exhaust the decision deadline. Conversely, a short status exchange can suppress a large redundant record. These are communication effects; changes in the evidence contract, finalizer, or runtime gate must be evaluated separately.

The framework therefore imposes four comparison invariants: evidence requirements remain fixed within a communication comparison; supporting and conflicting records share one validation boundary; evidence location does not confer finalization authority; and runtime outcomes are not credited to pre-action verification. Section V-B carries these invariants into the reporting checklist.

IV. FRAMEWORK CHECKS AND COMMUNICATION STUDY

A. Questions and Setup

This section checks the framework’s internal distinctions and implementation consistency, and then studies communica-

tion choices under controlled assumptions. It does not evaluate deployment effectiveness or runtime safety. The analysis asks three questions. *RQ1*: Can the framework distinguish same-proposal episodes that differ in proposal binding, freshness, conflict, or finalization authority? *RQ2*: Does the evaluator expose evidence transfer and evidence coordination as distinct observable effects? *RQ3*: Can One-way and Feedback be compared while evidence requirements, finalizer, and evidence quality are held fixed?

The synthetic setup has two evidence items and two endpoints. Each endpoint observation is missing, invalid, supporting, or conflicting; forward and reverse delivery each succeed or fail. The same 4096 weighted states are enumerated once and reused across all four communication cases. Records follow the common validation path, freshness is rechecked at decision time, and the finalizer is fixed within each comparison. The episode begins after proposal formation and ends at authorized finalization. The common 10-ms processing time is included, and the baseline deadline is 33 ms. Section IV-B operationalizes the cases motivated by Table I.

Unconditional coverage is the probability of timely Accept over all proposal episodes. Selective error is the ground-truth-unsupported mass among timely accepts. Expected evidence payload counts transmitted evidence-record bytes over all proposal episodes, including episodes that later Reject or Abstain. These metrics describe the declared finite model; they are not measured robot, wireless, or safety results. Headers, acknowledgments, retransmissions, channel occupancy, airtime, and energy are not represented by the payload field.

B. Framework Checks

RQ1 – structural distinctions. The artifact executes 11 checks and passes all 11. Complete admissible support with no conflict returns Accept; an admissible conflict returns Reject even before support is complete; and incomplete support returns Abstain. Expired, untrusted, binding-mismatched, calibration-revoked, and excessively uncertain conflicts do not veto. Finally, complete evidence at an unauthorized endpoint does not produce authorized finalization. These checks operationalize the distinctions in Table I; they test declared framework logic, not the physical correctness of a sensor claim.

RQ2 – transfer versus coordination. With timing and expiry removed for this structural check, sender-finalized Feedback uniquely reaches a complete admissible support set in 64 enumerated states. Under the declared synthetic state distribution, these states carry 8.3% probability mass. This is an evidence-transfer effect, not an estimate of real episode frequency. Under receiver finalization and the declared One-way fallback, the pre-timing reachability mismatch between One-way and Feedback is 0 states. Feedback instead reduces expected evidence payload by 6.49 KiB per proposal episode, an evidence-coordination effect without enlarging the reachable evidence set; its extra message can nevertheless change the timely result. At the ordinary timing baseline, the generated expected payloads are 9.80 KiB for One-way, 8.45 KiB for sender-finalized Feedback, and 3.31 KiB for receiver-finalized

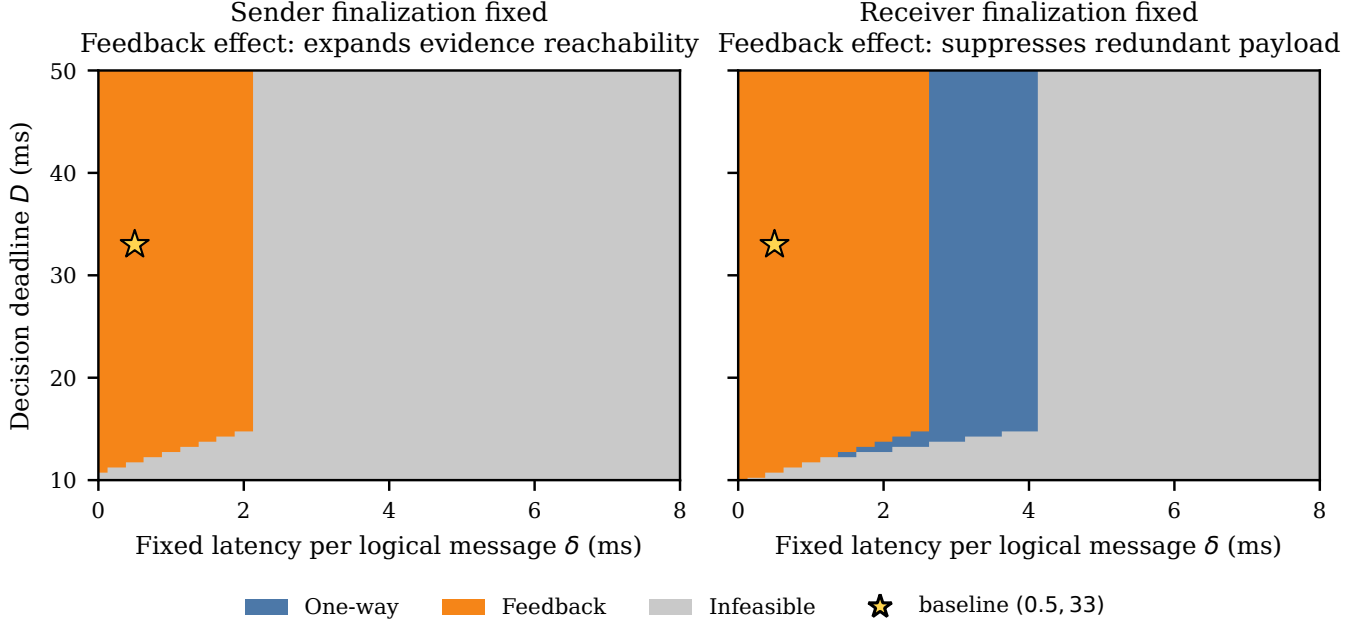


Fig. 3. Controlled communication study with fixed evidence requirements, authorized finalizer, and evidence-quality model. The axes vary fixed latency per logical message and the decision deadline. Feasibility requires at least 40% unconditional coverage and at most 0.75% selective error; feasible alternatives are ordered by expected evidence payload and then logical-message count, with exact ties assigned to One-way. The star marks the baseline.

Feedback; the matched receiver-finalized One-way value is 9.80 KiB.

C. Communication Study

RQ3 – requirement-preserving communication choice. Figure 3 holds the two evidence requirements, authorized finalizer, evidence-quality model, record sizes, link rate, processing time, initial ages, and lifetime fixed. It varies only fixed latency per logical message and the decision deadline. A candidate is feasible only if unconditional coverage is at least 40% and selective error is at most 0.75%. Among feasible candidates, the declared rule minimizes expected evidence payload, then logical-message count, with an exact tie assigned to One-way.

The gray region is infeasible under the declared requirements. In the orange region, Feedback is the feasible lower-payload choice. In the blue region, Feedback loses too much deadline or freshness budget, or both alternatives are feasible and One-way wins the message-count tie. At the marked baseline both panels select Feedback, but for different reasons: transfer expands the evidence reachable at a sender finalizer, whereas coordination suppresses redundant payload before receiver finalization. The boundaries are a controlled synthetic result produced by the framework, not the paper’s main novelty. Neither interaction dominates across the declared timing regimes.

V. DISCUSSION AND FUTURE DIRECTIONS

A. Research Directions

Evidence-gap detection separates deterministic checking from AI assistance. Contract checks identify missing, invalid, expired, or binding-mismatched records; authorization checks

find an unavailable finalizer; admissible conflict yields Reject. Embodied multimodal models integrate language and sensor inputs [16]; uncertainty-aware planners seek help under ambiguity [18]. Within our framework, they could propose evidence needs or clarifications; checks still distinguish evidence gaps, authority gaps, and conflicts. Neither confidence nor diagnosis is an evidence record or replaces checks.

Active acquisition and representation maps camera, LiDAR, contact, peer, human, or radio sensing to named requirements under freshness, uncertainty, and deadlines. Interactive agents select visual, acoustic, haptic, and proprioceptive information-gathering actions [19]; ISAC and perceptive mobile networks jointly support radio communication and environmental sensing [30], [31]. Within our framework, observations qualify as evidence only after requirement mapping and validation of proposal binding, provenance, freshness, uncertainty, and source. Semantic and task-oriented communication support task-relevant representation and delivery [5], [6]. The representation must additionally preserve the claim relation and those validation attributes.

Closed-loop re-inference and runtime integration could build on embodied multimodal models, LLM planners, and world-model or digital-twin architectures that exemplify reasoning, replanning, and state-dependent revision [4], [16], [17]. Within our framework, such engines could use validated evidence records to revise a proposal. Revision creates a new binding, may select another application-approved template, and requires revalidation. The runtime gate remains downstream and unevaluated. AI agents may diagnose and plan, but the evidence contract, common validator, authorization policy, and runtime gate remain explicit decision boundaries.

TABLE II
MINIMUM REPORTING CHECKLIST FOR POST-SEMANTIC COMPARISONS.

Layer	Report	Confounding avoided
Evidence	Requirements; support/conflict semantics; record source, modality, freshness, provenance, uncertainty, and proposal binding.	A change in what counts as sufficient is not mislabeled as a delivery gain.
Delivery	One-way or Feedback; feedback function; success model; expected evidence payload per proposal episode; logical-message count; latency and deadline accounting.	Payload suppression is not confused with total on-wire or energy reduction.
Finalization	Authorized finalizer; evidence-assembly location; credential and handoff assumptions.	Evidence location is not mistaken for action authority.
Runtime	Gate inputs, re-evaluation time, stop or backup action, and the execution-observation boundary.	Verifier Accept is not presented as a physical-safety result.

B. Reporting Guidelines

Table II gives the minimum fields needed to preserve the separations in Section III when comparing communication designs.

First, each field should be declared before comparison as fixed, varied, or unobserved. Evidence requirements, finalizer, freshness, and deadline are application inputs; the communication policy is the experimental factor.

Second, an episode-level trace should link the proposal, admissible and rejected records, messages, verifier outcome, authorized finalizer, and any observed runtime result. Stable record identifiers and validation outcomes can preserve this structure without publishing sensitive payloads.

Third, aggregate metrics should use explicit and common denominators. Unconditional coverage and expected payload use all proposal episodes; selective error uses timely Accept episodes; runtime rates use only episodes in which execution was observed. This accounting keeps the framework layers comparable without inventing unmeasured outcomes.

VI. CONCLUSION

A task-effective proposal is not yet a justified physical action. The verification gap is the mismatch between such a proposal and the valid, timely, proposal-bound evidence and authorization required at the permitted finalizer before action. Post-semantic communication makes explicit what must happen after a meaningful proposal is formed but before a physical action may proceed. The Post-Semantic Communication Framework makes the missing objects and responsibilities explicit through evidence requirements, proposal-bound records, conflict-first verification, authorized finalization, and a downstream runtime boundary. Evidence transfer and evidence coordination explain two different reasons for communication after proposal formation. The finite-state checks verify implementation consistency with the declared operational distinctions, while the controlled communication study demonstrates one requirement-preserving comparison of One-way and Feedback. The reporting guidelines carry the same separations into future studies. Runtime effectiveness and physical safety remain subjects for measured systems; runtime control stays downstream of verification and authorized finalization.

APPENDIX A

FORMAL OBSERVATIONS AND STRUCTURAL PROPERTIES

Observation 1 — Proposal-only representations can collapse distinct outcomes. Let $\pi(w)$ be a reduced proposal representation and let $a^*(w)$ be the required outcome. Suppose two physical or system states satisfy

$$\pi(w_1) = \pi(w_2), \quad a^*(w_1) \neq a^*(w_2). \quad (1)$$

If the outcome difference is caused by discarded proposal binding, decision-time freshness, admissible conflict, or finalization authority, no decision rule based only on $\pi(w)$ can be correct in both states.

Justification. For a proposal-only rule g , equality of the reduced representations gives

$$g(\pi(w_1)) = g(\pi(w_2)). \quad (2)$$

The required outcomes differ, so the common value must disagree with at least one state. This is a statement about the reduced representation; a richer semantic representation could carry the omitted fields. $\square$

Structural Property 1 — Conflict-first verifier consistency. Let $V_q(e, d, t)$ be the common validation predicate for evidence record e , proposal d , decision time t , and contract q . Define the admissible set as

$$E_q(d, t) = \{e : V_q(e, d, t) = 1\}. \quad (3)$$

The verifier

$$\Gamma_q(d, E_q) = \begin{cases} \text{Reject,} & \text{an admissible conflict exists,} \\ \text{Accept,} & \text{required support is complete,} \\ \text{Abstain,} & \text{otherwise} \end{cases} \quad (4)$$

satisfies three properties: inadmissible records neither support nor veto; admissible conflict precedes support completeness; and complete admissible support without conflict is accepted.

Argument. Validation first removes every record for which $V_q = 0$, establishing the first property. The first branch handles every remaining conflict before the support branch. If that branch is false and the required support is complete, the second branch applies; all residual cases enter the third branch. These exhaustive cases establish consistency with the declared verifier rule. They do not establish physical safety or deployment correctness. $\square$

Authorized finalization is a separate predicate. For endpoint i and declared authorization policy $\mathcal{A}$, an Accept result can authorize action only when

$$\Gamma_q = \text{Accept} \quad \text{and} \quad i \in \mathcal{A}. \quad (5)$$

APPENDIX B COMMUNICATION, FRESHNESS, AND AVAILABILITY MODEL

Fixed latency δ is charged once per logical message, evidence payload is serialized at rate R , and common pre-action processing T_p covers validation, assembly, and finalization. The ready times are

$$T_1(B_f) = \delta + 8B_f/R + T_p, \quad (6)$$

$$T_F(B_r, B_f) = 2\delta + 8(B_r + B_f)/R + T_p. \quad (7)$$

For initial record age A_e , lifetime L_e , and deadline D , a record used at finalization must satisfy

$$A_e + T_x \leq L_e, \quad T_x \leq D. \quad (8)$$

For one evidence item, let endpoint holding indicators have marginal probabilities q_S and q_R and Pearson correlation ρ . The evaluator uses

$$\Pr(H_S = 1, H_R = 1) = q_S q_R + \rho \sqrt{q_S(1 - q_S)q_R(1 - q_R)}. \quad (9)$$

The joint probability must lie within the Fréchet bounds; the remaining three Bernoulli cells follow from the marginals. This correlation applies across endpoints for each item, not between the two independent claim truths.

APPENDIX C FEEDBACK FUNCTION SEPARATION AND REQUIREMENT-BASED SELECTION

Structural Property 2 — Separation of feedback functions. Consider a fixed evidence contract and finalizer. For sender finalization, if a receiver holds a missing admissible record, the reverse transfer succeeds, and the record remains valid at decision time, Feedback can strictly enlarge the admissible evidence set reachable at the finalizer. For receiver finalization under a One-way fallback, coordination does not enlarge pre-timing evidence reachability relative to One-way, but successful status delivery can weakly reduce forward evidence payload when the receiver already holds a required record. Added latency or expiry may still change the timely outcome.

Argument. In the first case, let E_S be the sender-held admissible set and e_R a required record held only by the receiver. Successful transfer yields

$$E_S \subset E_S \cup \{e_R\}, \quad (10)$$

with strict inclusion; the added record may complete support. In the second case, One-way sends every sender-held admissible record. Delivered status lets the sender omit only records already available at the receiver; lost status invokes the same One-way set. The receiver's pre-timing union is therefore unchanged, while transmitted payload is no larger and is smaller whenever a positive-size redundant record is

omitted. Feedback has an extra logical message, so decision-time freshness and deadline tests can reverse the timely outcome. $\square$

For fixed finalization authority, let C_x , E_x , B_x , and N_x denote unconditional coverage, selective error, expected evidence payload per proposal episode, and logical-message count for interaction x . Feasibility requires

$$C_x \geq C_{\min}, \quad E_x \leq E_{\max}. \quad (11)$$

Among feasible candidates, the controlled illustration minimizes

$$(B_x, N_x, I_x) \quad (12)$$

lexicographically, where I_x resolves an exact tie in favor of One-way. This is an application-declared ordering rather than a general utility. Sender-finalized Feedback can improve C_x through transfer; receiver-finalized Feedback can reduce B_x through coordination. In either case, an added message can reduce timely coverage through reverse loss, latency, record ageing, or expiry.

APPENDIX D SYNTHETIC SETUP AND REPRODUCTION BOUNDARY

Table III records the fixed synthetic declarations. The per-claim truth prior and per-endpoint evidence availability happen to be 0.90 but are different quantities. The two truth bits are independent; availability for the same item is correlated across endpoints as defined in Appendix B.

TABLE III
SYNTHETIC BASELINE DECLARATIONS.

Parameter	Baseline
Required evidence	Visual clearance + Radio no-motion
Claim-truth prior p_{truth}	0.90 per claim
Claim-truth dependence	Independent
Evidence availability q	0.90 per endpoint/item
Conditional validity	0.99
Evidence-relation error	0.10
Endpoint correlation ρ	0.25
Delivery success (p_f, p_r)	(0.99, 0.99)
Records (B_V, B_R)	(10, 1) KiB
Link rate R	100 Mbit/s
Per-message latency δ	0.5 ms
Processing time T_p	10 ms
Decision deadline D	33 ms
Initial ages (A_V, A_R)	(20, 15) ms
Evidence TTL	35 ms

Exact enumeration assigns weights to two truth bits, four endpoint-record observations, and two delivery bits. Missing, invalid, supporting, and conflicting observations are evaluated on the same weighted state space for all four communication cases. Communication determines which validated records travel, charges their bytes, computes action-ready time, revalidates lifetime, and returns Accept, Reject, or Abstain. A separate authorization check determines whether Accept can be finalized. Exact enumeration removes Monte Carlo sampling noise only; it does not establish realism.

The 11 generated RQ1 checks are: complete support; incomplete support; an admissible Visual conflict; conflict before

support completeness; conflicts made inadmissible by invalid provenance, proposal-binding mismatch, expiry, revoked calibration, an untrusted source, or excessive uncertainty; and complete evidence at an unauthorized endpoint. The first four exercise the verifier's Accept, Abstain, and conflict-first Reject behavior; the next six verify that inadmissible conflicts do not veto; the last separates evidence sufficiency from the authorization policy. The generated CSV and JSON retain each check's name, expected outcome, observed verification and finalization outcomes, and pass flag.

The RQ2 structural calculation uses the same weighted states but removes deadline and lifetime truncation. This isolates whether interaction changes the admissible set that can reach a fixed finalizer before any timing penalty is applied. The sender-transfer metric sums the probability of states in which Feedback reaches complete support and One-way does not. The receiver reachability metric counts any One-way/Feedback outcome mismatch under the declared status-loss fallback. A separate unconditional expectation measures the evidence-record bytes suppressed by coordination. Timing is restored for the ordinary baseline values and for Figure 3.

For RQ3, each grid cell reuses the same truth, record-observation, and delivery population. Only the fixed latency per logical message and the deadline change. The generator recomputes timely Accept and selective error for both interactions, applies the common feasibility requirements, and then applies the declared payload/message ordering.

All reader-facing baseline values in Section IV are loaded from generated LaTeX macros. Generated artifacts and validation establish internal source-to-claim consistency within the declared finite model.

The model contains no measured wireless trace, queue, scheduler, retransmission model, calibrated sensor dataset, exercised authorization infrastructure, or measured runtime controller. Consequently, Figure 3 is a constructed design result rather than deployment evidence. The runtime gate is represented only as a downstream boundary.

The arXiv source package contains the code and generated artifacts used for this version, identified by the snapshot string `arxiv24-v25.1.0`. This version does not assert a public repository, tagged archival release, or Zenodo DOI. The source-bound snapshot does not establish calibration, certification, evidence-contract validity, sensor correctness, runtime effectiveness, or physical safety.

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